

Toward a Theory of Computation for Learned Transformer Machines: Tasks, Resources, Learnability, Composition, and Closure

Paper 0.1 — Position and Formal Framework

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Abstract

The dominant theoretical question about large language models is often phrased too coarsely: “what can a Transformer compute?” Classical expressivity results characterize idealized Transformers through formal languages and circuit classes, while empirical work usually asks whether a trained model succeeds on a benchmark. Between these views lies a missing object: the computational theory of a *learned* Transformer machine. A computation may be representable by an architecture yet rarely acquired by training; it may be learned on bounded instances yet fail to compose or close over instance depth. Moreover, the computational object in a modern application is usually compound:

$$\mathcal{M} = (F_\theta, C, S, I, \mathbb{T}, \mathcal{B}),$$

combining a learned controller, execution protocol, writable state, information interfaces, a tool algebra, and finite resource budgets. These are product-space coordinates, not a scalar “agent level.” We distinguish information sufficiency, representability, acquisition, execution reliability, efficiency, systematic generalization, and closure. Tasks are classified by required state and iteration; experiments estimate competence and acquisition frontiers rather than infer algorithms from endpoint accuracy. Papers 0.5–0.9 provide later empirical anchors, but none is required for the framework.

1 Why a learned-machine theory is needed

A fixed architecture can represent computations that gradient-based training rarely discovers. A trained model can fit finite examples without acquiring a reusable rule. A rule can be reusable over a bounded range without supporting closure. Conversely, autoregressive generation can repeatedly apply the same learned local transformation and thereby change the effective computational machine without changing the model weights.

We therefore separate:

$\text{representability} \neq \text{learnability} \neq \text{reliable acquisition} \neq \text{systematic executability} \neq \text{closure}.$

Classical Transformer theory provides essential upper and lower bounds under explicit architectural and precision assumptions. Formal-language work relates fixed-depth attention models to circuit classes such as AC^0 and TC^0 , while recent chain-of-thought theory shows that autoregressive intermediate computation can substantially expand expressivity. Our objective is complementary: characterize the *trained finite machine* and measure how its empirical computational frontier depends on architecture, training distribution, generated state, context, and external execution resources.

2 Basic objects

2.1 Task family

A task family is

$$\mathcal{T} = (\mathcal{X}, \mathcal{Y}, \mathcal{D}, \mathcal{R}),$$

where $\mathcal{D}$ is a generator/distribution and $\mathcal{R}$ is the declared structural prior that makes prediction possible.

Each instance has a complexity vector

$$\mathbf{C} = (S, K, B, V, N, M, \dots),$$

where S is atomic syntax/operator complexity, K required composition/proof depth, B branching/search width, V variable/state complexity, N nuisance context, and M memory/context requirement. Complexity is generally partially ordered, so measured Pareto competence regions are preferable to a single scalar difficulty.

2.2 Information sufficiency

Before studying an intelligent mechanism, determine whether the task is information-theoretically solvable. For query X , context D , and target Y :

$$I(Y; D \mid X) = H(Y \mid X) - H(Y \mid X, D).$$

For deterministic rule inference under declared prior $\mathcal{R}$, require

$$H(Y \mid X, D, \mathcal{R}) = 0.$$

An arbitrary unseen random mapping with insufficient demonstrations is not a reasoning problem: no oracle lacking additional information can exceed the Bayes ceiling.

Every task release should therefore include an information-validity record

$$\mathcal{I} = (H(Y \mid X), H(Y \mid X, D), I(Y; D \mid X), \mathcal{R}, A_{\text{Bayes}}, |\mathcal{H}_{X,D,\mathcal{R}}|, m_{\text{id}}),$$

where $\mathcal{H}_{X,D,\mathcal{R}}$ is the set of hypotheses consistent with the query, demonstrations, and declared prior, and m_{id} is the minimum context needed for identifiability. Entropies are distribution-dependent quantities, not properties of a serialization alone. If the generator is deterministic only after conditioning on $\mathcal{R}$, that conditioning must be stated rather than smuggled into an oracle.

Proposition 1 (Information impossibility). *Under zero-one loss, if $H(Y \mid X, D, \mathcal{R}) > 0$, then no predictor measurable with respect to $(X, D, \mathcal{R})$ is universally correct. Its optimal expected accuracy is*

$$A_{\text{Bayes}} = \mathbb{E}_{X,D,\mathcal{R}} \max_y P(Y = y \mid X, D, \mathcal{R}) < 1$$

whenever the conditional distribution is non-degenerate on a set of positive measure.

Pointwise, the best decision selects the most probable conditional label; positive conditional entropy entails ambiguity on at least one event of positive measure. This separates model failure from task underdetermination.

2.3 Controller hierarchy and compound machines

The controller coordinate has four operational levels. At C_0 , a fixed pass computes $Y = F_\theta(X)$. At C_1 , autoregression computes $y_t = F_\theta(X, y_{<t})$; generated tokens form serialized writable state. At C_2 , the state language is structured into fields such as CURRENT, TARGET, COUNTER, STACK, SUBSTITUTION, or PROOF-STATE. At C_3 , an external loop repeatedly inspects state, selects an action or tool, incorporates its result, and decides whether to continue. A practical “agent” is typically a C_3 protocol, not automatically a new formal class.

Controller sophistication is only one coordinate. We define the compound learned machine

Definition 1 (Compound learned machine).

$$\mathcal{M} = (F_\theta, C, S, I, \mathbb{T}, \mathcal{B}),$$

where F_θ is the learned Transformer/controller; C is its execution protocol; S is the writable-state interface; $I : q \mapsto D$ supplies external information; $\mathbb{T} = \{T_1, \dots, T_m\}$ is a set of partial external operations $T_i : \mathcal{X}_i \rightarrow \mathcal{Y}_i$; and $\mathcal{B}$ bounds context, calls, tokens, time, and memory. For acquisition process Π ,

$$A(\tau, \mathbf{C}; F_\theta, C, S, I, \mathbb{T}, \mathcal{B}, \Pi), \quad P_{\text{acquire}} = P_\Pi[A \geq \gamma].$$

This product description replaces a scalar tool/agent ordering. Autoregression and structured traces alter C and S ; retrieval primarily alters I ; a local graph edge, a closure oracle, and Python alter $\mathbb{T}$ in categorically different ways. Every comparison should record

$$\Delta \mathcal{M} = (\Delta C, \Delta S, \Delta I, \Delta \mathbb{T}, \Delta \mathcal{B}, \Delta \Pi),$$

so an easier prompt, a stronger primitive, more execution time, and better training are not all called “augmentation.”

3 Six computational axes

3.1 Axis 1: atomic syntax and local operator learning

What is the most complex single-step relation a model of size (L, W, H) learns reliably? A rough ladder is

$$\begin{aligned} &\text{association} \rightarrow \text{n-gram} \rightarrow \text{unary} \rightarrow \text{binary} \rightarrow \text{Boolean rule} \\ &\rightarrow \text{predicate+variables} \rightarrow \text{nested functor} \rightarrow \text{type dispatch} \rightarrow \text{rewrite}. \end{aligned}$$

Define acquisition probability

$$P_{\text{acquire}}(\tau; M, \mathcal{D}, T) = P_s[A_s(\tau) \geq \gamma]$$

and coverage over the intended legal instance set. The falsifiable resource question is whether a controlled change in width, heads, layers, data, or budget moves this boundary across seeds. A single successful seed establishes neither learnability nor reliable acquisition.

3.2 Axis 2: internal one-pass composition

Assume local operator f is competent. How much composition can one forward pass implement?

$$x \mapsto f(x) \mapsto f^2(x) \mapsto \dots \mapsto f^K(x).$$

Define

$$K_{\text{internal}}(M, \tau) = \max\{K : A(f^K; M) \geq \gamma\}.$$

A larger frontier does not prove serial layer-by-layer execution. Required dependency radius, operator count, proof depth, branching, nuisance, and raw length must be crossed rather than conflated. Ordered intermediate decoding and causal transfer are required before an $L \times K$ interaction is interpreted as serial internal computation.

3.3 Axis 3: autoregressive multi-token composition

Autoregression exposes recurrence:

$$s_{t+1} = F_\theta(X, s_{\leq t}).$$

For a parent relation, a root trajectory may be serialized as

$$C \rightarrow B \rightarrow A \rightarrow R.$$

Self-attention can read the entire generated history, so the state may be a tuple, proof prefix, substitution, or counter. Define K_{AR} and compare it directly with K_{internal} . Report local transition accuracy, stopping accuracy, total-sequence success, generated steps, and train-short/test-long behavior. This distinguishes a competent transition with accumulated error from a failed recurrence controller.

3.4 Axis 4: closure

Distinguish:

$$\forall K < \infty \exists M_K$$

(bounded family solvability), from a fixed-model tested range

$$\exists M \forall K \leq K_{\text{max}}^{\text{test}},$$

and from true algorithmic closure

$$\exists M \forall K < \infty.$$

The last cannot be established empirically, but closure signatures can be tested by train-short/test-long extrapolation, computation length scaling with instance complexity, local-transition reuse, and termination generalization. Finite context creates a separate practical memory boundary.

Definition 2 (Budgeted closure signature). *For a fixed trained machine m and explicit context/step budget C_{max} , define*

$$\Sigma_{m, C_{\text{max}}} = (K_{\text{train}}, K_{\text{test}}, K_{\text{pass}}, \rho_{\text{step}}, \rho_{\text{stop}}, g(K), f_{\text{fail}}),$$

where K_{pass} is the largest tested complexity passing the multi-seed gate, ρ_{step} and ρ_{stop} are local-transition and stopping reliability, $g(K)$ is observed computation-length scaling, and f_{fail} is the first classified failure. This is a finite empirical object, not algorithmic closure.

3.5 Axis 5: in-context information

ICL changes what facts, rules, or parameters are available without changing weights. Separate fact acquisition from procedure acquisition. Contextual tasks should be classified as copy, constraint completion, finite task selection, parametric rule induction, or compositional/procedural induction. Holding X fixed while varying D is essential. Even when $I(Y; D \mid X) > 0$, controls must distinguish copying, elimination, task selection, parameter estimation, and construction of a reusable procedure.

3.6 Axis 6: harness and tool augmentation

External execution may expand representable computation, improve acquisition, stabilize execution, or merely save resources. These are different claims. A proper factorial comparison crosses controller C_0 – C_3 with tool semantics rather than placing “tools” above autoregression on one ladder. Without controller-matched and tool-matched baselines, delegated computation can be mistaken for neural acquisition.

4 Information interfaces and tool algebra

An information interface $I(q) = D$ changes the evidence available to the controller. Its characteristic contribution is a conditional-entropy reduction $H(Y \mid X, D) < H(Y \mid X)$, not necessarily a stronger inference procedure. Basic retrieval-augmented generation is therefore approximated by

$$q \mapsto I(q) = D \mapsto F_\theta(q, D),$$

or “external information selection plus in-context computation.” Ranking, query decomposition, iterative search, and filtering may add control or computation, but must be classified separately from retrieval itself.

The tool coordinate is semantic. We use: T_0 , information lookup such as **retrieve** or **read**; T_1 , a local transition such as **parent**, **neighbor**, or **successor**; T_2 , a deterministic transformation such as arithmetic, sorting, simplification, or unification; T_3 , search or closure such as **root**, **shortestPath**, or recursive SQL; and T_4 , a programmable executor such as Python, shell, SQL, or Cypher. A single T_2 or T_3 call may hide many elementary steps. A T_4 interpreter may be universal under suitable language and resource assumptions, yet this says nothing about learned program synthesis, reliability, description length, or efficiency.

Let $\langle \mathbb{T} \rangle_C^g$ denote computations obtainable by finite, budgeted composition of $\mathbb{T}$ under controller C . A C_3 controller with **parent** can represent repeated parent^k if it also represents state update and termination. Supplying **root** instead moves the task-defining closure inside the tool. Supplying an interpreter $T_{\text{int}}(p, x) = p(x)$ moves the learned problem further toward program synthesis and call reliability. Computability equivalence between two interfaces therefore does not imply equal acquisition probability or cost.

Tool/information	C_0 fixed pass	C_1 autoregressive	C_2 structured state	C_3 iterative control
None	one-shot prediction	ordinary generation; CoT	typed scratchpad	repeated model/environment loop
T_0 lookup T_1 local	basic RAG context one external step	retrieve then generate serialized local calls	cited/evidence state explicit current/counter	iterative RAG GraphRAG traversal; parent iteration
T_2 transform	delegated transform	calculator/PAL call	typed expression/result	CAS or nonrecursive SQL agent
T_3 closure	delegated answer	closure API call	explicit query/result	shortest-path or recursive-query agent
T_4 interpreter	execute supplied program	generate then execute	program plus execution state	code/SQL/Cypher agent

Table 1: Controller $\times$ tool/information product space. Entries are examples, not formal equivalence classes; state, budget, and training remain separate coordinates.

4.1 Delegation and four kinds of gain

For benchmark function f , a tool pipeline can be written conceptually as

$$f = g \circ T \circ h,$$

where h serializes arguments, T performs an external operation, and g integrates the result. With $T = \text{parent}$, root finding still requires controller-side iteration and termination. With $T = \text{root}$, the benchmark is directly delegated. Every tool result should identify where the task-defining computation occurs.

Four gains must be reported separately: *computational gain*, when a resource makes a computation representable under stated base-machine constraints; *acquisition gain*, when it raises P_{acquire} although both machines can represent the procedure; *execution-reliability gain*, when it raises $P_{\text{exec}}(\text{success} \mid \text{acquired})$; and *efficiency gain*, when it reduces examples, generated steps, errors, latency, memory, or model size. Endpoint accuracy alone cannot identify which occurred.

As an accounting identity rather than an independence model,

$$\begin{aligned} P(\text{success}) &= P(\text{information sufficient}) \\ &\quad \times P(\text{procedure acquired} \mid \text{information}) \\ &\quad \times P(\text{execution succeeds} \mid \text{acquired, information}). \end{aligned}$$

Tool execution further factors into action selection, argument correctness, tool success, result integration, and termination. These conditional events are the natural failure taxonomy for Paper 0.9.

Proposition 2 (Direct delegation). *If tool T computes benchmark target f exactly and a controller reliably serializes the input and interprets the output, success establishes acquisition of the call-and-integrate interface, not neural acquisition of f .*

The distinction is empirical: argument formation and result use may remain difficult, but the defining computation occurs in T .

Proposition 3 (Budgeted composition of a local primitive). *If controller C reliably maintains state, invokes local primitive T , and applies a correct termination predicate, then $T^k(x)$ is representable for every finite k whose trace fits $\mathcal{B}$.*

This is a representability statement, not a guarantee that Π acquires the controller or that errors remain bounded.

4.2 Common systems are compound configurations

RAG, GraphRAG, calculators, and agents are application patterns rather than single computational classes. GraphRAG is heterogeneous: $\text{neighbor}(v)$ is T_1 , while community aggregation is a substantial T_2 -like transform. Shortest-path and common-ancestor APIs are T_3 ; Cypher is a T_4 -like programmable query interface. Microsoft GraphRAG combines graph construction, network analysis, retrieval, prompting, and summarization; its product name does not determine one row of the algebra [16].

System	Controller	Information	Tool algebra	Added resource	Task-defining computation
One-shot / CoT Basic / iterative RAG	C_0/C_1 $C_0, C_1/C_3$	prompt retrieved documents	none T_0	pass / serial tokens external evidence	learned controller retrieval plus in-context use
Local GraphRAG	C_3	graph	T_1 neighbor	graph access, iteration	controller traversal
Search GraphRAG	C_3	graph	T_3 path/search	delegated closure	graph engine
Calculator / CAS	C_1/C_3	prompt	T_2	arithmetic/symbolic engine	often external engine
SQL agent	C_3	database	T_2/T_4	query language/engine	split; recursive SQL can close
Code interpreter	C_3	files/state	T_4	loops, recursion, memory	synthesized program + runtime
ReAct / multi-tool	C_3	observations	mixed	action-observation loop	depends on selected primitive

Table 2: System mapping. Calculator serialization, CAS operations, SQL planning, and code generation remain learned tasks even when execution is external.

5 Task classes by required state

Class	Required computation	Example	Minimal resource in principle	Empirically helpful resource
A: local	$x \mapsto f(x)$	parent	C_0 or one T_1 call	width/data or reliable call
B: fixed composition	$f^k(x)$	grandparent	finite C_0 depth or bounded trace	C_1/C_2 state
C: termination closure	iterate to predicate	root	recurrence + stop	$C_3 + T_1$; T_3 delegates
D: variable count	$(x, k) \mapsto (f(x), k - 1)$	ancestor $_k$	counter/state + recurrence	$C_2/C_3 + T_1$
E: multi-state	update several pointers	paired traversal	tuple state + recurrence	typed C_2 fields
F: search	frontier/visited set	proof search	memory + branching control	C_3 ; T_3 delegates search
G: binding	environment σ	unification	substitution state	C_2 or T_2 unifier
H: proof state	repeated rule choice	theorem proving	structured state + control	C_3 plus local rules

Table 3: Task-state progression. “Minimal” is a representability statement under explicit budgets; “helpful” is an empirical hypothesis, and closure/search tools may delegate the task.

6 A learned-computation phase diagram

For machine class $\mathcal{M}$, task family τ , complexity $\mathbf{C}$, and training process Π , define

$$A(\tau, \mathbf{C}; \mathcal{M}, \Pi), \quad P_{\text{acquire}}(\tau, \mathbf{C}; \mathcal{M}, \Pi).$$

Here Π is a distribution over initialization s_θ , minibatch/order s_b , generator s_g , and optimizer/hyperparameters h , not a single lucky run:

$$P_{\text{acquire}} = \Pr_{s_\theta, s_b, s_g, h} \{A \geq \gamma, A_{\text{ctrl}} < \gamma_{\text{ctrl}}\}.$$

For a tested domain $\Omega_{\tau, \mathbf{C}}$, coverage is the measured mass or fraction on which the gate holds. The internal and autoregressive frontiers K_{internal} and K_{AR} are task- and protocol-specific projections of a generally non-rectangular competence region. A complete report distinguishes Bayes solvability, representability, learnability, reliable acquisition, coverage, composition frontier, recurrence frontier, and closure signature.

Let Q be the latent event that a reusable procedure was acquired and S the event of evaluation success. Then

$$P(S) = P(Q)P(S | Q) + P(\neg Q)P(S | \neg Q).$$

The shorthand $P(S) \approx P_{\text{acquire}}P(S | Q)$ is valid only when shortcut success in the $\neg Q$ class is negligible. Because Q is not directly observed, controls and interventions identify a useful mixture model rather than a unique psychological state.

Proposition 4 (Local-error accumulation). *If an execution requires T transitions and, conditional on every previously correct state, each transition is correct independently with probability $1 - \epsilon$, then exact trajectory success is $(1 - \epsilon)^T \leq e^{-\epsilon T}$.*

Independence is a diagnostic baseline, not a default model: correlated errors, recovery, verification, or history-dependent attention can produce slower or faster decay. Reporting both local and sequence accuracy tests which regime applies.

Proposition 5 (Finite evidence does not establish closure). *For any finite tested set $K \leq K_{\text{test}}$, there exist a closed procedure and a bounded lookup machine that agree on every observation and diverge for some $K > K_{\text{test}}$.*

Construct the lookup machine to copy the closed procedure on the finite tested domain and return a fixed wrong answer outside it. Thus empirical evidence can support closure signatures and falsify particular bounded hypotheses, but cannot prove the universal quantifier.

No empirical resource monotonicity is assumed. Although a larger architecture may contain a smaller computation representationally, its optimizer, implicit bias, attention geometry, and finite-data basin differ. Hence $P_{\text{acquire}}(L + 1) \geq P_{\text{acquire}}(L)$ is an empirical hypothesis, not an axiom.

Failure labels should include:

Bayes-impossible, missing fact, operator, composition, recurrence, counter/control, termination, search, context overflow, accumulated local error, acquisition-basin failure.

7 Relation to classical computation theory

Classical automata theory classifies languages and machines using memory, state transition, stack access, and recursion. Circuit complexity classifies computations by size, depth, fan-in, and gate type. The analogies below are conditional descriptions of resources, not identities between neural architectures and classical machines.

A C_0 fixed Transformer is naturally compared with a bounded circuit only after specifying hard versus soft attention, numerical precision, positional encoding, uniformity, sequence-length scaling, and whether depth and size are fixed or grow. AC^0/TC^0 -style statements established for one variant do not transfer automatically to another [1, 2, 10, 5]. A C_1 autoregressive Transformer adds serial computational time through generated tokens; it is better described as a learned sequential transducer with readable serialized history than as a finite-state automaton absent an explicit finite-state restriction. Li et al.’s separation result makes this resource change precise under their assumptions [6].

A C_2 scratchpad can serialize counters, stacks, substitutions, proof states, registers, or tape-like configurations. These analogies identify the state a task requires; they do not show that the architecture acquired a counter machine, pushdown automaton, register machine, or Turing machine. A C_3 tool controller resembles an instruction-set- or oracle-augmented machine, but “oracle” suppresses real tool cost and failure. A T_4 interpreter can provide general program execution under finite resources, making pure computability too coarse: learned program description, synthesis probability, execution reliability, and budget dominate practical behavior.

The present program adds an empirical learned-machine dimension. A construction proving that a Transformer *can* implement an operation establishes representability, not that SGD will find it from a finite generator. Turing completeness in principle does not imply reliable algorithm acquisition. Conversely, bounded empirical failure does not establish architectural impossibility.

The desired bridge is

machine resources + task structure + training distribution
 $\longrightarrow$ acquisition and execution frontier

8 Related work

Table 4 locates the learned-machine layer relative to representative theory. The labels summarize theorem assumptions rather than asserting equivalence between models.

Work	Machine	Assumptions emphasized	Result type	Gap addressed here
Hahn (2020)	fixed attention	bounded depth; formal-language setting	limitation/sensitivity	training and finite acquisition
Hao et al. (2022)	fixed hard attention	circuit assumptions, precision	AC^0 -related bounds	empirical competence regions
Merrill et al. (2022)	saturated attention	asymptotic precision/size assumptions	threshold-circuit characterization	finite learned frontier
Pérez et al. (2021)	recurrent/encoded Transformer	explicit precision/encoding assumptions	Turing-completeness construction	probability of learning construction
Strobl et al. (2024)	multiple variants	positions, precision, uniformity	expressivity survey	common reporting schema
Li et al. (2024)	autoregressive CoT	serial intermediate tokens	expressivity separation	acquired finite K_{AR}
Joshi et al. (2025)	autoregressive learner	time-invariant generator	learning/sample analysis	seed and generator acquisition
Bavandpour et al. (2025)	hard-attention CoT	scratchpad assumptions	length lower bounds	measured closure signatures
Huang et al. (2025)	trained recognizer	regular-language distributions	training dynamics	broader task/resource phase diagram

Table 4: Representative related-work matrix. Precision, positional assumptions, and uniformity must be read from each theorem; no row transfers automatically to the finite models studied here.

Formal-language and circuit-complexity analyses have established limitations and capabilities of fixed-depth Transformer variants under explicit assumptions [1, 2]. Strobl et al. survey this literature and emphasize that conclusions depend critically on precision, positional encodings, attention definitions, and uniformity [5]. Universality and Turing-completeness constructions establish strong representability under richer assumptions [4, 3], but do not answer finite-data acquisition probability.

Recent chain-of-thought theory is especially relevant. Li et al. show that intermediate autoregressive steps can enable inherently serial computations inaccessible to shallow one-pass models under their assumptions [6]. Joshi et al. formalize learning with an iterated time-invariant next-token generator and analyze sample/computational complexity [7]. Bavandpour et al. derive lower bounds on scratchpad length for algorithmic problems in hard-attention Transformers [8]. Huang et al. study training dynamics for regular-language recognition [9]. These works motivate treating generated tokens as a computational resource rather than merely explanatory prose.

Compound-system work supplies concrete configurations. Retrieval-augmented generation adds a learned retriever and retrieved evidence to generation [11]. MRKL routes queries among neural and symbolic modules [12]; Toolformer studies self-supervised acquisition of tool calls [13]; program-aided language models delegate execution of generated programs [14]; and ReAct interleaves learned reasoning traces, actions, and observations [15]. Their endpoints should be reanalyzed through Table 1: what information, state, primitive semantics, control, and budget changed, and was the observed gain computational, acquisitional, reliable, or efficient?

Mechanistic evidence answers another layer. Induction-head work demonstrates circuit emergence and links it to in-context behavior [17]; residual-stream decompositions provide a vocabulary for attributing component updates [18]; and causal tracing tests necessity by controlled activation replacement rather than by geometry alone [19]. Circuit existence, causal necessity, acquisition dynamics, systematic generalization, and length/closure extrapolation are progressively stronger and should not be collapsed into one claim of “understanding.”

Our emphasis is different: measure learned finite-machine frontiers, acquisition probability, information sufficiency, and transitions between one-pass, recurrent, contextual, and harnessed execution in tiny models where causal inspection is feasible.

9 Implications for Paper 0.9

Paper 0.9 should not compare an undifferentiated “model” with an “agent.” Its primary factorial design crosses controller—one-token C_0 , free autoregression C_1 , structured state C_2 , iterative control C_3 —with tool—none, local T_1 , deterministic T_2 , closure/search T_3 , and programmable T_4 . The first contrast should be none versus a local primitive such as `parent(x)`. That leaves iteration and termination with the controller and therefore tests recurrence acquisition. A `root(x)` or shortest-path API is an upper delegation control; it tests calling and integration, not learned closure.

For every trajectory Paper 0.9 should record information sufficiency, procedure acquisition, action selection, argument correctness, tool success, result integration, and termination. Conditional local accuracy and exact trajectory accuracy test the independent-error baseline; correlated failure and self-correction must be measured rather than assumed. Results should report controller-by-tool competence, call and token budgets, execution cost, and where the benchmark-defining computation occurs. This factorization connects failure modes to the compound machine instead of assigning them vaguely to “agency.”

10 Distributed cognition and heterogeneous computational intelligence

The compound-machine formalism admits a broader, secondary interpretation: useful computation need not be localized in one model. Distributed cognition has long analyzed cognition at the level of people, representations, artifacts, and environment [20]. This systems stance is related to, but weaker than, the extended-mind thesis of Clark and Chalmers [21]. Nothing here requires a metaphysical claim that a database, API, or notebook literally constitutes a person’s mind. The claim is operational: heterogeneous components can participate in one stateful computation, and the location of task-relevant work can be measured across their interfaces.

10.1 A heterogeneous computational graph

Extend a compound machine to a directed graph

$$\mathcal{G} = (V, E), \quad N_i : (X_i, S_i) \longrightarrow (Y_i, S'_i), \quad v_i \in V,$$

where an edge $e_{ij} : Y_i \rightarrow X_j$ is a communication channel or callable interface. Nodes may be learned models, humans, calculators, databases, retrievers, graph engines, theorem provers, interpreters, APIs, or microservices. This shared node notation asserts interface participation, not internal equivalence. A human and a deterministic service differ radically in adaptation, grounding, memory, failure, and mechanism even when both accept a request and return a result.

For each node, retain a vector rather than an “intelligence score”:

$$\chi_i = (\mathcal{F}_i, c_i, \ell_i, r_i, a_i, m_i, \kappa_i),$$

where $\mathcal{F}_i$ is its supported function family, c_i cost, ℓ_i latency, r_i reliability, a_i adaptability/generalization, m_i state capacity, and κ_i its iteration or closure characteristics. A calculator is narrow and exact; an LLM is broad and adaptive but probabilistic; a database exposes a narrow query interface over large persistent state; a human is broadly adaptive and grounded but slow, costly, and working-memory limited. The vector makes complementarity visible without erasing mechanism.

For goal g , a controller selects a possibly branching, stateful trajectory

$$\pi_g = (N_{i_1}, N_{i_2}, \dots, N_{i_T}), \quad A(g; \mathcal{G}, \pi_g, \mathcal{B}).$$

Writing $F_{\pi_g} = N_{i_T} \circ \dots \circ N_{i_1}$ is useful shorthand, but real trajectories exchange messages, revisit nodes, branch on observations, and mutate state. Routing is itself a learned or programmed problem. A cautious design objective is to choose π under accuracy, cost, latency, reliability, and resource constraints; no single scalar utility is assumed here.

An orchestrating model need not implement every sub-operation internally, but it must recognize a subproblem, select a node, parameterize its interface, integrate the result, verify it when needed, update state, and continue or terminate. Corresponding conditional quantities include

$$P_{\text{select}}, \quad P_{\text{parameterize}}, \quad P_{\text{integrate}}, \quad P_{\text{verify}}, \quad P_{\text{continue}}.$$

A perfect solver is useless behind an interface the controller cannot invoke. These terms refine the earlier decomposition into action choice, argument formation, and result integration. They make orchestration an empirical capability rather than a synonym for intelligence.

10.2 Reachable composition and distributed closure

Network capability is neither the maximum nor the simple union of node capabilities:

$$\mathcal{F}_{\mathcal{G}} \neq \bigcup_i \mathcal{F}_i, \quad \mathcal{F}_{\mathcal{G}} \subseteq \langle \mathcal{F}_1, \dots, \mathcal{F}_n \rangle_{\text{routing, state, budget}}.$$

The left inequality allows a routed composition that no node realizes alone; the right inclusion emphasizes that inaccessible interfaces, bad serialization, missing state, unreliable routing, or insufficient budget can prevent even individually available operations from composing. This is exactly the controller–state–information–tool algebra lifted from one compound machine to a network: each node exposes a local transition or compound submachine, while edges and routing determine reachable compositions.

A component may have only a finite measured frontier K_i , while the network exhibits a larger *system-level* or *distributed closure signature*. For example, an LLM that cannot reliably execute f^{1000} internally may synthesize a loop and invoke a T_4 interpreter. Then $K_{\text{LLM}} < K_{\text{LLM}+\text{interpreter}}$ over the measured budget. The closure belongs to the routed system, and the LLM’s learned contribution may be program synthesis and result integration rather than iteration. As before, finite context, calls, time, memory, and interpreter limits prohibit an unqualified claim of mathematical closure.

10.3 Real systems and non-equivalent nodes

Humans have long externalized memory and computation into speech, writing, notation, books, maps, calculators, computers, and networks. This is not a strict evolutionary ladder; it is a recurring pattern of persistent representation and specialized delegation. LLM-centered systems add a potentially useful node type: an adaptive semantic interface and probabilistic controller between rigid typed services. This flexibility complements, rather than dominates, the high reliability of narrow deterministic components.

System	Representative nodes/route	Distributed contribution	Principal failure boundary
RAG assistant	human $\rightarrow$ LLM $\leftrightarrow$ retriever/store	evidence access plus semantic integration	retrieval sufficiency, citation/use
Graph assistant	LLM $\leftrightarrow$ graph store/API	local traversal or delegated search by API semantics	query choice, closure delegation
Calculator/CAS	LLM $\rightarrow$ arithmetic/symbolic engine $\rightarrow$ LLM	semantic dispatch plus exact transform	serialization and interpretation
Coding agent	human $\rightarrow$ LLM $\rightarrow$ editor/compiler/tests/repository	deterministic feedback on generated actions	patch validity, test coverage, stopping
Scientific workflow	human, LLM, Python, simulator, database, statistics	heterogeneous modeling and verification loop	model choice, data validity, review
Enterprise workflow	human, LLM, APIs, SQL, business rules	semantic routing among typed services	authorization, schema, policy, escalation

Table 5: Examples of heterogeneous goal execution. No row implies that humans, learned models, and services are mechanistically or cognitively equivalent.

The system abstraction can make it less practically important that one node internally realize every sub-operation, but it makes component science more important, not less: system capability depends on the actual frontier, calibration, and failure distribution of each node. A trajectory such as human $\rightarrow$ coding model $\rightarrow$ compiler $\rightarrow$ tests $\rightarrow$ repository $\rightarrow$ model $\rightarrow$ human distributes proposal, execution, checking, persistent state, and final judgment across non-equivalent components.

10.4 Reliability, verification, and qualified analogies

For a path with conditional node/call reliabilities r_t , $\prod_t r_t$ is only an independent-error baseline. Real systems contain correlated failures, retries, redundancy, self-correction, and escalation. A *verification topology* can improve system reliability by routing generated code through a compiler and tests, a proof through a theorem checker, an API request through a schema validator, arithmetic through a calculator, or high-risk output through a human. Verification adds calls and latency, and is only useful when checks cover the relevant failure modes.

Classical distributed systems supply helpful analogies—message passing, state, scheduling, latency, failure, retries, consistency, and resource allocation—but learned nodes are distribution-sensitive, semantic interfaces may be underspecified, and decomposition/routing may itself be inferred. Likewise, multi-agent LLM systems are the homogeneous special case $N_i = \text{LLM}_i$; adding agents does not by itself create a stronger class. Node semantics and communication topology determine the change. Distributed problem solving and multi-agent research provide the established computational context for such coordination [22]; Paper 0.1 contributes a connection to learned acquisition frontiers, tool algebra, reliability, and measured closure, not the broad novelty claim that cognition can be distributed.

The practical unit of analysis may therefore be a heterogeneous computational network rather than an isolated model. That perspective remains downstream of the core theory: first characterize local machines and interfaces; then ask which compositions the graph can acquire, route, verify, and execute under finite resources.

11 Constructive representability calibration

The framework separates representability from acquisition; fixed-weight witnesses make that distinction concrete. We manually specify three micro-Transformers with identity residuals, causal finite-softmax attention, ReLU FF blocks, zero positional embeddings, and no LayerNorm or dropout. No weight is trained or fitted. Every matrix and canonical forward trace is recorded, and each declared tiny legal domain is exhaustively enumerated.

Task / data	Primary machine	(L, H, d)	Full-domain result	Component controls	Constructive interpretation
Successor / weights	FF-only	(1, 0, 4)	3/3; margin 1	SA 0/3; SA+FF 3/3	exact local linear/ReLU map
Pair lookup / context	SA-only	(1, 1, 9)	18/18; margin > .9999995	FF 6/18 ties; SA+FF 18/18	finite-softmax key/value transport
Grandparent / context	SA-only	(2, 1, 20)	2/2; margin > .9999995	FF 0/2; SA+FF 2/2; one-layer SA 0/2	bounded two-hop residual composition

Table 6: Milestone-1 manually weighted witnesses. “SA” and “FF” columns are encoding-relative controls, not universal architectural lower bounds.

For associative lookup, the desired scaled score is 16 and three other final-query scores are zero, giving finite probability $e^{16}/(e^{16} + 3) = .99999962395$. Grandparent uses the same finite selector twice: layer one writes the parent into a dedicated state subspace and layer two uses that state to retrieve the grandparent. The failed one-layer control transports only one hop. These constructions establish that the specified architectures can implement the functions under atomic pair/edge encodings. They establish neither minimality, uniqueness, robustness beyond the legal domain, nor that SGD will acquire these weights or mechanisms. Full matrices and forward traces are in the companion manual; exhaustive rows and matrix CSVs are under `results/manual_witnesses/`.

12 Empirical anchors from the series

These anchors occur after the framework because none is needed for its definitions or propositions. They are bounded observations, not universal claims.

Paper 0.5: one-pass refinement. Across three controlled generators, depth improves predictive-class organization, usually non-monotonically. Signal grows faster than fluctuation; absolute margin variance does not contract. Local-attention reachability is necessary but insufficient, and head effects are distributed and non-additive. This anchors Axes 1–2 while falsifying simple monotone-depth, contraction, and stable-head-role stories.

Paper 0.6: competence and allocation. Bounded structured attribute composition is a positive generator-specific result, but the original depth-general predicate phase passes no three-seed gate. A later capacity diagnostic finds that depth 8 makes `isAncestor` seed-stable on easy and medium tiers, but not hard; four other predicates remain far below competence. At twice the training budget, the complete baseline also passes easy and medium while failing hard, so optimization moves the boundary and weakens a depth-only explanation. Depth eight is near-perfect at 2x for two observed seeds, but its third seed is missing and the central gate is unevaluable. This remains a finite membership frontier, not general predicate closure.

Paper 0.8: information versus acquisition. The D4 permutation-completion task is information-sufficient: under its declared prior, context reduces target uncertainty from $\log_2 3$ to zero. Yet none of five Pareto-small candidates passes all three seeds. In contextual-copy calibration, one competent checkpoint exhibits pair binding and later value routing, while fresh replications miss the 95% three-seed gate. At 1,024 episodes, seeds 11 and 23 reach 100%, whereas seed 37 moves from 52.1% at nominal budget to 81.2% at 2x and 60.4% at a targeted 4x run. This directly motivates separating information sufficiency, physical mechanism after acquisition, and reliable acquisition.

Papers 0.7, 0.85, and 0.9: pending gates. Paper 0.7 now has exact CPU-validated propositional and unification generators, but no formal-model frontier is yet a finding. Paper 0.85 will compare one-token and unconstrained multi-token machines; Paper 0.9 will compare scratchpad, proto-tool, and harness transitions. Their planned comparisons are hypotheses, not empirical anchors.

13 Core falsifiable conjectures

- C1.** Reliable acquisition of a local operator does not imply reliable acquisition of its finite compositions.
- C2.** Many iterative task families exhibit a finite one-pass composition frontier for a fixed trained Transformer.
- C3.** Multi-token state generation substantially expands the measured composition frontier for tasks admitting a reusable local transition.
- C4.** After local competence, long-recurrence failure increasingly reflects state representation, termination, counters, and accumulated error.
- C5.** Supplying facts/rules in context separates factual memory failure from procedural failure.
- C6.** For some tasks, explicit harnesses greatly improve acquisition/execution reliability without changing what an ideal autoregressive Transformer could represent in principle.

14 Open problems

The framework sharpens eight questions. Which local operators have high acquisition probability at a fixed scale and distribution? Which resource interactions move one-pass composition frontiers? When does autoregressive state produce a stable train-short/test-long signature rather than a longer bounded repertoire? Which tasks genuinely require counters, multiple state variables, stacks, substitutions, or visited-state memory under a specified interface? Does context provide missing facts, choose among stored tasks, estimate parameters, or construct a transition? When does a harness enlarge the semantic machine rather than stabilize an already representable trajectory? Can empirical acquisition boundaries be connected to circuit or automata classes without discarding precision and uniformity assumptions? Finally, what generator and curriculum properties favor a reusable transition over memorized finite coverage?

15 Conclusion

A theory of computation for LLMs should distinguish what a model can represent, what training can reliably acquire, what one pass can compose, what autoregressive recurrence can close, what context contributes as information, and where external execution performs the task. The computational object in a modern application is not the language model alone: it is a learned controller embedded in a state architecture, supplied with information interfaces and an external instruction algebra under finite resource bounds. Classical computability characterizes what this compound machine can represent in principle. A theory of learned machines must additionally characterize which procedures training discovers, with what probability, over which complexity frontier, at what cost, and with what execution reliability.

References

- [1] M. Hahn. Theoretical Limitations of Self-Attention in Neural Sequence Models. *TACL*, 2020.
- [2] Y. Hao, D. Angluin, and R. Frank. Formal Language Recognition by Hard Attention Transformers: Perspectives from Circuit Complexity. *TACL*, 2022.
- [3] J. Pérez, P. Barceló, and J. Marinkovic. Attention is Turing Complete. *JMLR*, 2021.
- [4] C. Yun et al. Are Transformers Universal Approximators of Sequence-to-Sequence Functions? *ICLR*, 2020.
- [5] L. Strobl, W. Merrill, G. Weiss, D. Chiang, and D. Angluin. What Formal Languages Can Transformers Express? A Survey. *TACL*, 2024.
- [6] Z. Li, H. Liu, D. Zhou, and T. Ma. Chain of Thought Empowers Transformers to Solve Inherently Serial Problems. *ICLR*, 2024.

- [7] N. Joshi et al. A Theory of Learning with Autoregressive Chain of Thought. *COLT*, 2025.
- [8] A. A. Bavandpour et al. Lower Bounds for Chain-of-Thought Reasoning in Hard-Attention Transformers. *ICML*, 2025.
- [9] R. Huang, Y. Liang, and J. Yang. How Transformers Learn Regular Language Recognition: A Theoretical Study on Training Dynamics and Implicit Bias. *ICML*, 2025.
- [10] W. Merrill, A. Sabharwal, and N. A. Smith. Saturated Transformers are Constant-Depth Threshold Circuits. *TACL*, 2022.
- [11] P. Lewis et al. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. *NeurIPS*, 2020.
- [12] E. Karpas et al. MRKL Systems: A Modular, Neuro-Symbolic Architecture That Combines Large Language Models, External Knowledge Sources and Discrete Reasoning. arXiv:2205.00445, 2022.
- [13] T. Schick et al. Toolformer: Language Models Can Teach Themselves to Use Tools. *NeurIPS*, 2023.
- [14] L. Gao et al. PAL: Program-Aided Language Models. *ICML*, 2023.
- [15] S. Yao et al. ReAct: Synergizing Reasoning and Acting in Language Models. *ICLR*, 2023.
- [16] D. Edge et al. From Local to Global: A Graph RAG Approach to Query-Focused Summarization. arXiv:2404.16130, 2024.
- [17] C. Olsson et al. In-Context Learning and Induction Heads. *Transformer Circuits Thread*, 2022.
- [18] N. Elhage et al. A Mathematical Framework for Transformer Circuits. *Transformer Circuits Thread*, 2021.
- [19] K. Meng et al. Locating and Editing Factual Associations in GPT. *NeurIPS*, 2022.
- [20] E. Hutchins. *Cognition in the Wild*. MIT Press, 1995.
- [21] A. Clark and D. Chalmers. The Extended Mind. *Analysis*, 58(1):7–19, 1998.
- [22] M. Wooldridge. *An Introduction to MultiAgent Systems*, second edition. Wiley, 2009.